BLM5121 – Web Mining Term Project

Spring 2026

1. Project Overview

In this project, you will design and implement a web mining application that addresses a real-world problem in one of the three core web mining domains: Web Content Mining, Web Structure Mining, or Web Usage Mining. The project requires you to compare three distinct methods or algorithms applied to the same problem, evaluate their performance quantitatively, and present results through a graphical user interface.

The project requires you to deeply understand three core web mining concepts:

- Web Content Mining: extracting useful information from the content of web pages (text, images, multimedia).
- Web Structure Mining: analyzing the hyperlink structure of the web to discover patterns (e.g., PageRank, HITS).
- Web Usage Mining: discovering usage patterns from web server logs and user interaction data.

2. System Architecture

Your application must consist of the following interconnected components:

2.1 Data Acquisition Module

A module responsible for managing and preprocessing the selected dataset:

- Preprocess raw data: cleaning, tokenization, normalization, and feature extraction.
- Split data into training/test sets where applicable.
- Store processed data in an appropriate local format (CSV, JSON, or database).

2.2 Algorithm Comparison Module (3 Methods)

You will implement and compare three different methods/algorithms that address the same chosen problem. All three methods must be applied to the same dataset and evaluated under identical conditions:

Method 1 – Baseline Algorithm

- Implement a classical or widely-known approach relevant to the chosen domain.
- Record execution time, resource usage, and output quality metrics.

Method 2 – Alternative Algorithm

- Implement a second approach with different computational characteristics.
- Document key differences in assumptions and complexity.

Method 3 – Advanced / Proposed Algorithm

- Implement a third method (e.g., machine learning-based, graph-based, hybrid).
- Compare directly against Methods 1 and 2 using the same evaluation metrics.

2.3 Graphical User Interface (GUI)

Your application must include a GUI (web-based, desktop, or notebook-based) with the following three screens:

Screen 1 – Dataset Manager

- Display dataset statistics: number of records, feature summary, and class distributions.
- Allow basic filtering and exploration of data samples.

Screen 2 – Algorithm Runner & Comparator

- Select and run individual algorithms or all three simultaneously.
- Display real-time progress and status during execution.
- Show side-by-side performance metrics (accuracy, F1, runtime, etc.).
- Visualize results with charts and comparison tables.

Screen 3 – Results & Analysis

- Present ranked comparison of all three methods.
- Display output examples (e.g., top-ranked pages, discovered clusters, usage patterns).
- Export results and plots for inclusion in the final report.

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3. Dataset Requirements

You are free to choose your own dataset, but it must be publicly available and suitable for a web mining problem. The dataset must meet the following criteria:

- Must be selected from a public repository such as Kaggle, UCI ML Repository, or similar.
- Must be directly relevant to the chosen web mining domain.
- Provide the source URL in your proposal and final report.

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4. Performance Analysis

You must conduct and report a thorough performance analysis including:

- Execution time of each algorithm for different dataset sizes (e.g., 10%, 50%, 100% of dataset).
- Accuracy, Precision, Recall, and F1-Score (or domain-appropriate metrics) for each method.
- Memory usage and scalability characteristics.
- Statistical significance testing where applicable.
- Visual comparison: bar charts, ROC curves, confusion matrices, or equivalent.

All results must be presented as tables and graphs in the final report.

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5. Step 1 – Submit a Proposal

Write a 1-page project proposal. The proposal is free-format but cannot exceed one page. Use your best judgement regarding margins and font size. The proposal should be concise, concrete, focused and to the point. It must answer the following basic questions:

- **Why?** Motivate your idea: is it interesting, important, and relevant to web mining?
- **What?** Exactly what problem do you propose to solve, and which three algorithms will you compare?
- **How?** State your evaluation procedure, metrics, and dataset.
- **When?** Provide a realistic milestone-based timetable from now until the submission deadline.

The proposal must also include:

- Dataset description: name, source URL, format, and record count.
- Justification: why this dataset and problem are suitable for web mining.
- Student information: full name and student ID (individual project).

Proposal Due Date: Hard copy must be submitted in class during the 8th week of the semester.

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6. Step 2 – Final Submission

Implement your complete system and write a project report (minimum 2 pages, maximum 5 pages, excluding figures and references). The report must cover:

- Motivation, goals, and implementation environment (language, libraries, OS).
- Description of the selected dataset and preprocessing steps.
- Detailed explanation of each of the three algorithms (assumptions, complexity, parameters).

- Use case scenario with screenshots of all three GUI screens.
- System architecture diagram showing the data flow from input to output.
- Performance evaluation results with tables and charts.
- Discussion: which algorithm performed best and why? What are the trade-offs?
- Technical challenges encountered and how they were resolved.
- Conclusions and future work.

6.1 Submission Instructions

- Submit all source files and the report as a single .zip file.
- Filename format: TP_StudentID.zip
- Send to: blm5121.web.mining@gmail.com
- Email subject: [TermProject] StudentID – Full Name

Final Submission Deadline: One day before the last lecture of the semester, by 11:59 PM (midnight).

6.2 Late Submission Policy

Late submissions will incur a grading penalty. Projects submitted after the deadline will be graded with a deduction proportional to the delay. No exceptions will be made without prior approval from the instructor.

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7. Tentative Grading Criteria

Component	Points	Notes
Dataset Preprocessing	15	Cleaning, normalization, feature extraction
Algorithm 1 – Implementation	15	Correct implementation + results
Algorithm 2 – Implementation	15	Correct implementation + results
Algorithm 3 – Implementation	15	Correct implementation + results
Comparative Analysis	15	Fair comparison, same conditions
GUI (3 screens)	10	All screens functional
Performance Evaluation	10	Charts and tables required
Report Quality	5	2–5 pages, well structured
TOTAL	**100**	